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Acid-Base Promoted Dehydrogenation Coupling of Ethanol on Supported Ag Particles

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ABSTRACT

Dehydrogenation coupling of ethanol to n-butanol has received much attention due to the wide application of n-butanol. Highly selective butanol production from ethanol remains a challenge due to competitive reactions in the dehydrogenation and/or condensation and great difficulty in ethanol dehydrogenation. This work reports a synergistic catalysis between layered double hydroxide (LDO), an acidic-basic support, and supported Ag particles for the dehydrogenation coupling of ethanol to n-butanol. A selectivity of up to 77 % toward n-butanol with ethanol conversion of 23.2 %, amongst impressive levels reported till now, has been achieved at 350 °C, 0.1 MPa. A selectivity of up to 60 % toward n-butanol with ethanol conversion of 45 %, the highest selectivity at an ethanol conversion of > 45 % reported till now, has been achieved at 250 °C, 2 MPa. A synergistic catalysis between the acid-base sites of LDO surface and supported Ag

particles has been revealed to contribute the excellent butanol selectivity: basic sites promote adsorption and subsequent dehydrogenation of ethanol on Ag particles; in addition to the promotion from Ag, the following aldol condensation is boosted by acid-base cooperation.

1. INTRODUCTION

Acceptorless dehydrogenation coupling or hydrogen transfer reactions of alcoholic molecules have been attracting much interest in recent years due to their excellent step-economy and high atom-efficiency, providing a direct approach for the construction of C-C bonds.¹⁻⁶ Among the dehydrogenation coupling of alcohols, n-butanol production from bioethanol has currently been receiving great attention⁷⁻¹¹, in view of the wide applications of butanol as a crucial bulk chemical¹² and the advantages of butanol over bioethanol as a sustainable fuel^{13, 14}. Industrially, n-butanol is currently produced from hydroformylation¹⁵ of the fossil-based propylene followed by hydrogenation of the resulting butyraldehyde, or produced by the fermentation of starchy bodies^{12, 16, 17}. As alternative, ethanol coupling offers an economical and sustainable route for n-butanol production.⁷⁻¹¹ Alcohol coupling has been proposed to proceed by a Guerbet reaction⁷⁻¹¹ or a direct condensation mechanism¹⁸⁻²⁰. In the direct condensation, the β -H of ethanol molecule is activated and then condensate with another ethanol through dehydration. In the Guerbet reaction, the C-C formation is accomplished through an aldol-condensation route, which involves the dehydrogenation of ethanol to aldehyde, the aldol-type condensation of aldehyde, and the hydrogenation through a hydrogen transfer.

Since the first report on barium ethoxide catalyzed ethanol coupling to n-butanol by Guerbet²¹ in 1901, both homogeneous^{10, 11} and heterogeneous^{7, 11} catalytic strategies have been developed. A

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2
3 ruthenium-bis(diphenylphosphanyl)methane complex has been reported to afford a 94 %
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5 selectivity to n-butanol with > 20 % conversion in the presence of EtONa.²² A > 99 % selectivity
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7 at 37 % conversions has been obtained on [Ir] catalyst coupled with nickel or copper hydroxide
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9 complexes.²³ Non-noble metal catalyst, Mn-complex has been developed for coupling of ethanol
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11 in the presence of EtONa, affording a selectivity of 92 % to n-butanol.²⁴ Even though a better
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13 performance has been achieved in homogeneous systems, heterogeneous catalytic coupling of
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15 liquid ethanol to n-butanol arouses interest, with supported Au, Cu, Pd, or Ru as catalysts²⁵⁻²⁷ in
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17 a batch reactor. A 70 % selectivity to n-butanol at 21 % conversion has been achieved on
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19 supported Au catalyst at 275 °C under supercritical conditions.²⁷ Due to its potential large-scale
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21 industrial application, continuous catalytic coupling of ethanol to n-butanol in heterogeneous
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23 catalytic systems has been attracting continuous attention and comprehensively investigated in
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25 the last decades. In 1931, the dehydrogenation coupling of ethanol was first observed on
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27 mixtures of MgO, Al₂O₃, and CuO_x in conjunction with H₂.²⁸ Since then, heterogeneous catalysts
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29 with acidic and/or basic sites (i.e. oxide or mixed oxides,²⁹⁻³⁴ alkali metal-modified zeolites,³⁵
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31 and hydroxyapatite³⁶⁻⁴²) have been widely developed for the continuous conversion of ethanol to
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33 butanol. About 33 % selectivity to n-butanol at 56 % ethanol conversion has been achieved on
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35 MgO at 723 K and 1 bar³². 76.3 % selectivity to n-butanol at 14.7 % conversion or 86.4 %
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37 selectivity at 11.3 % conversion has been produced on Ca-P HAP³⁸ or Sr-P HAP⁴¹ at 573 K and
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39 atmospheric pressure. Various transition metals have also been explored to enhance the ethanol
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41 dehydrogenation under mild conditions, thus improving ethanol conversion and decreasing
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43 reaction temperature.⁴³⁻⁴⁷ Incorporation of Ni to MgAlO afforded a selectivity of 55.2 % to n-
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45 butanol with an ethanol conversion of 18.7 %.⁴³ An average selectivity of 40 % towards n-
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47 butanol with ethanol conversions of 44.0-46.2 % has been achieved over Cu-CeO₂/AC at 523 K
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and 2 MPa N₂.⁴⁴ A selectivity of 48.6% toward n-butanol with an ethanol conversion of 49.8 % has been also achieved over UiO-66-encapsulated Pd at 523 K and 2 MPa N₂.⁴⁵

Supported Ag catalysts have been extensively applied in oxidant-free dehydrogenation of alcohols.⁴⁸⁻⁵⁰ The oxidant-free dehydrogenation is exactly needed in the coupling of ethanol to butanol through the Guerbet process. So this work employs supported Ag as a catalyst. To promote the catalysis of Ag particle, Mg and Al-containing layered double oxides (MgAl-LDO), derived from Mg and Al-containing layered double hydroxides (MgAl-LDHs) by calcination has been applied as a support because of its acidic and basic bifunctionalities.⁵¹⁻⁵⁴ To modulate the concentration of acid and basic sites, the Mg/Al ratio has been tailored. A selectivity of up to 76.6 % to n-butanol with a site time conversion (STC) of 2200.8 h⁻¹ has been achieved on Ag particles supported on MgAl-LDO (Ag/MgAl-LDO) with a Mg/Al ratio of 2. The high selectivity to n-butanol on Ag/MgAl-LDO is primarily ascribed to the synergies of acid-base sites coexisting on the surface of MgAl-LDO and uniformly dispersed Ag particles, where basic sites promote the adsorption and subsequent dehydrogenation of ethanol on Ag particles and the acid-base cooperation promotes the aldol condensation of dehydrogenated intermediate to form C-C bonds.

2. EXPERIMENTAL SECTION

2.1 Materials

Analytical-pure chemicals in this work, such as the nitrates mentioned below, sodium hydroxide, sodium carbonate, and ammonia, were purchased from Sinopharm. Chromatographically pure ethanol was purchased from Aladdin.

2.2 Preparation

MgAl-LDHs, Mg(OH)₂ or AlO(OH): MgAl-LDHs with varied Mg/Al molar ratios, Mg(OH)₂, and AlO(OH) were prepared following the procedure reported in our previous work⁵⁵.

Ag⁺ loaded MgAl-LDHs, Mg(OH)₂, or AlO(OH) : Ag⁺/MgAl-LDHs, Ag⁺/Mg(OH)₂, and Ag⁺/AlO(OH) were prepared by incipient wetness impregnation of MgAl-LDHs, Mg(OH)₂, and AlO(OH) with AgNO₃ solution, following the procedure reported in our previous work⁵⁵.

Ag loaded MgAl-LDO, Al₂O₃, and MgO: The prepared Ag⁺/MgAl-LDHs was calcined at 500 °C for 1 h under N₂ atmosphere with a heating rate of 5 °C min⁻¹, affording Ag/Mg₂Al-LDO, Ag/Mg₃Al-LDO, and Ag/Mg₄Al-LDO with a Ag loading of 0.68 % in each case. For a better comparison, Ag/Al₂O₃ or Ag/MgO was prepared by calcination of Ag⁺/AlO(OH) and Ag⁺/Mg(OH)₂ under N₂ atmosphere under a calcination temperature reduced to 400 °C while with the reduction time prolonged to 3 h, to ensure a similar size distribution of Ag particles in Ag/Al₂O₃ or Ag/MgO to that in Ag/MgAl-LDO. The Ag loadings were determined by ICP technique between 0.73 wt% and 0.64 wt%.

Mg₄Al-LDO, Al₂O₃, or MgO: Mg₄Al-LDO was prepared by calcination of MgAl-LDHs with Mg/Al ratio of 4.15 under N₂ atmosphere at 500 °C for 1 h, and Al₂O₃ and MgO were prepared by calcination of AlO(OH) and Mg(OH)₂ under N₂ atmosphere at 400 °C for 3 h. The

temperature was programmed from room temperature to 400 or 500 °C in a heating rate of 5 °C min⁻¹.

2.3 Characterizations

Powder X-ray diffraction (XRD) patterns were recorded with a range of 3°-80° at 10° min⁻¹ on a Shimadzu XRD-6000 diffractometer using a $\lambda = 0.1541$ nm radiation. ICP-ES was performed for the quantitative analysis for Ag, Mg and Al on a Shimadzu ICPS-7500. Brunauer-Emmett-Teller (BET) method was applied to analyze the specific surface area on a Micromeritics ASAP 2460. TEM and HRTEM images were taken on a Tecnai G² F30 at 300 kV. The Ag dispersion was determined by hydrogen–oxygen titration (HOT) on a Micrometric ChemiSorb 2920 with a thermal conductivity detector (TCD) following the procedure reported in our group⁵⁶ and calculated as follows:

$$D_{Ag} = \frac{N_{surface}}{N_{total}} = \frac{2 \times \text{amount of consumed } H_2}{N_{total}(\text{by ICP})} \times 100 \% \quad (1).$$

The surface basicity was measured by CO₂-thermal programmed desorption (TPD) on the Micrometric ChemiSorb 2920 following the procedure reported in our previous work^{55, 57}. The sample was pre-treated at 400 °C for 1 h under 40 mL min⁻¹ of He.

The pyridine-, acetaldehyde-, and ethanol-FT-IR spectra were recorded on an iS50 FT-IR (NICOLET) spectrometer following the procedure reported in our previous work^{55, 57}. For acetaldehyde adsorption, the spectrum for background was recorded at 10 °C under vacuum. Then acetaldehyde vapor was introduced to the system until equilibrium was reached. The system was purged with Ar until the spectra showed no change and then the spectrum was recorded.

2.4 Catalytic test

The coupling of ethanol were performed in a fixed-bed reactor with a stainless steel tubular reactor (10 mm external diameter, 38 cm length) equipped with an on-line gas chromatography (GC). Typically, 0.5 g of sieved catalyst (20-40 mesh) was loaded into the reactor and the remaining was filled with quartz sand (20-40 mesh). Prior to the reaction, the catalyst was pretreated in situ with N₂ (40 mL min⁻¹) at 400 °C for 3 h. Ethanol vapor and N₂ were mixed into the reaction system after the reactor was cooled to 350 °C. The N₂ gas flow was set to 60 mL min⁻¹. Before admission of ethanol into the reaction system, ethanol was vaporized at 150 °C in a vaporizing chamber. The chromatographically pure ethanol was pumped into the vaporizing chamber at 50 µL min⁻¹ using a high-performance liquid chromatography pump. The pipeline behind the reactor was heated to keep at 200 °C. The products were identified by GC-MS (Agilent 7200 Q-TOF GC/MS) and analyzed quantitatively by GC (Shimadzu, 2014C) with a flame ionization detector (FID) and a GSBP-INOWAX column (30 m, 0.25 mm inner diameter). The ethanol conversion and products selectivity were calculated as follows:

$$\text{Con. \%} = \frac{F_{in} - F_{unreacted}}{F_{in}} \times 100 \% \quad (4),$$

$$\text{Sel. \%} = \frac{C_{mol\ of\ specific\ product}}{C_{mol\ of\ products}} \times 100 \quad (5),$$

$$C_{balance} \% = \frac{C_{mol\ of\ products}}{C_{mol\ of\ conversed\ ethanol}} \times 100 \% \quad (6),$$

$$\text{STC} = \frac{F_{in} \times \text{Con.}}{n_{surface\ Ag}} \quad (7),$$

$$\text{Sel.\% of dehydrogenation} = 100 \% - \text{sel. \% for ethyl ether} \quad (8),$$

$$\text{Sel. \% of C - C coupling} = \sum \text{sel. \% for } C_4 \text{ and } C_4 + \text{ products} \quad (9),$$

wherein, F_{in} and $F_{unreacted}$ are the moles of initial ethanol and the unreacted ethanol; C mol is the mole number of carbon in the products and converted ethanol; STC is the site time conversion of ethanol and yield of n-butanol production. For comparison, the catalytic experiments were also performed at 250 °C and 2 MPa, following the procedure reported in literature ^{44, 45}.

3. RESULTS AND DISCUSSION

Figure 1 shows the XRD patterns of as-prepared hydroxides, Ag loaded hydroxides, and Ag loaded oxides. The characteristic reflections of hydrotalcite-like structure are clearly observed in the XRD patterns of Mg_xAl -LDHs (**Figure 1A, a-c**). The characteristic reflections of bohmite structure for $AlO(OH)$ (JCPDS: 21-1307) (**Figure 1A, d**) and brucite structure for $Mg(OH)_2$ (JCPDS: 44-1482) (**Figure 1A, e**) are also clearly. In each case, no reflections for other phases are observed (**Figure 1A**). No phase transformation is observed after impregnation with $AgNO_3$ solution (**Figure 1B**). By calcination of Ag loaded hydroxides under N_2 , the resulting Ag/ Mg_xAl -LDO (**Figure 1C, a-c**) display both of Mg-Al mixed oxide (Mg_xAl -LDO) phase (JCPDS: 45-0946) and metallic Ag phase (JCPDS: 04-0783). The characteristic reflections of Al_2O_3 (JCPDS: 04-0880) (**Figure 1C, d**) and MgO (**Figure 1C, e**) can be observed. Also the reflections for Ag phase were clearly for Ag/ Al_2O_3 (**Figure 1C, d**) and Ag/MgO (**Figure 1C, e**). The Ag loading was determined by ICP technique between 0.73 wt% and 0.64 wt%.

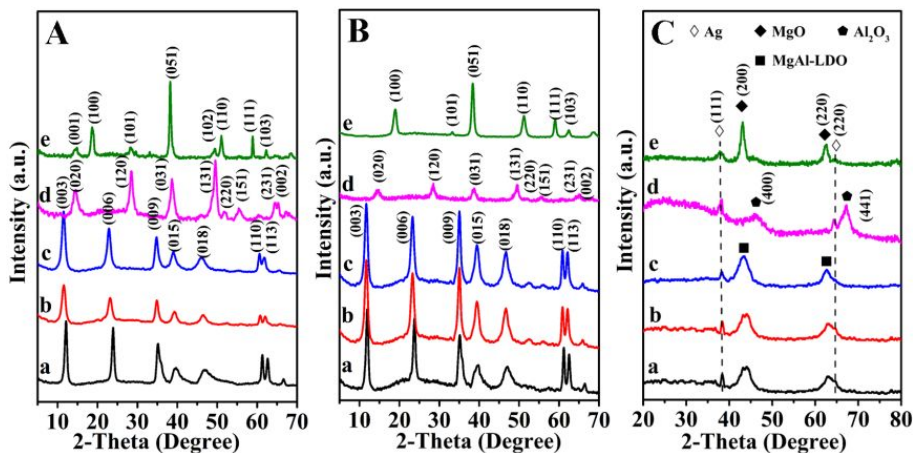


Figure 1. XRD patterns of hydroxides before (A) and after (B) Ag⁺ loading: (a) Mg_{2.23}Al-LDHs, (b) Mg_{3.17}Al-LDHs, (c) Mg_{4.15}Al-LDHs, (d) AlO(OH), and (e) Mg(OH)₂. (C) XRD patterns of (a) 0.68 wt% Ag/Mg_{2.23}Al-LDO, (b) 0.64 wt% Ag/Mg_{3.17}Al-LDO, (c) 0.65 wt% Ag/Mg_{4.15}Al-LDO, (d) 0.70 wt% Ag/Al₂O₃, and (e) 0.73 wt% Ag/MgO, prepared by calcination of (B).

In the TEM and HRTEM images for Ag loaded oxides (**Figure 2**), Ag particles disperse in a narrow distribution of particle size with the maximum at 7-9 nm, with (111) facet exposed in each case. The Ag dispersion has been determined as 15.4 % to 17.4 % for the Ag loaded oxides. Ag/MgAl-LDO or Ag/Al₂O₃ show similar BET surface area (57-86 m²/g) while Ag/MgO shows higher BET surface area (174.4 m²/g) (**Table 1**).

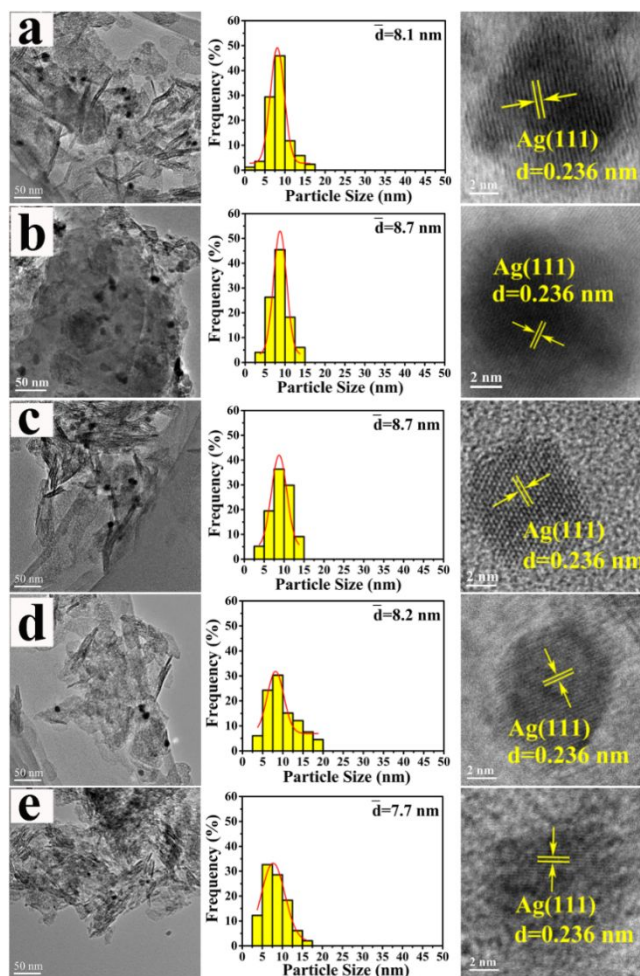


Figure 2. TEM images, size distribution, and HRTEM images of metallic Ag nanoparticle for (a) Ag/Mg₂Al-LDO, (b) Ag/Mg₃Al-LDO, (c) Ag/Mg₄Al-LDO, (d) Ag/Al₂O₃, and (e) Ag/MgO.

The surface basicity and acidity of Ag loaded oxides have been determined by CO₂-TPD and pyridine-FTIR, respectively. On Ag loaded LDO, a broad CO₂ desorption is observed between 50 and 450 °C (**Figure 3A, a-c**), which can be deconvoluted into three contributions identified⁵⁸⁻⁶⁰ as the adsorption of CO₂ on weak basic sites at 110-140 °C, medium-strong basic sites at 180~260 °C, and strong basic sites at > 280 °C. A higher proportion of medium-strong basic sites has been detected on Ag/Mg₃Al-LDO while the proportion for weak, medium-strong, and strong base sites on Ag/Mg₂Al-LDO and Ag/Mg₄Al-LDO (**Table 1**). Quantitatively, the total

amount and density of base sites decline with increasing Mg/Al molar ratio despite a difference in the ratio of medium-strong to strong base sites. For Ag/Al₂O₃ sample, almost no medium-strong and strong base sites but only a small amount of weak base sites are detected. For Ag/MgO sample, the amount of basic sites (0.61 mmol/g) is higher than that for Ag/Mg_xAl-LDO while the density of basic sites is less due to its higher BET surface area. Without Ag loaded, Mg₄Al-LDO possesses similar basicity to Ag/Mg₄Al-LDO, indicating that Ag loading has no marked influence on the basic sites. As shown in **Figure 3B**, only the adsorption on Lewis acid sites⁶¹ is observed. The total amount of the acid sites shows no visible change while the density of acid sites decreases with increasing Mg/Al molar ratio (**Table 1**). For Ag/Al₂O₃, visibly higher amount and density of acidic sites have been detected. For Ag/MgO, only trace of acid sites is detected. On Mg₄Al-LDO, similar acid sites are detected as on Ag/Mg₄Al-LDO. The ratio of acid to base sites increases with increasing Mg/Al molar ratio for Ag/MgAl-LDO, but all around 0.5 (from 0.41 to 0.69). The acid/base ratio for Ag/Al₂O₃ is much higher and the acid/base ratio for Ag/MgO is much lower.

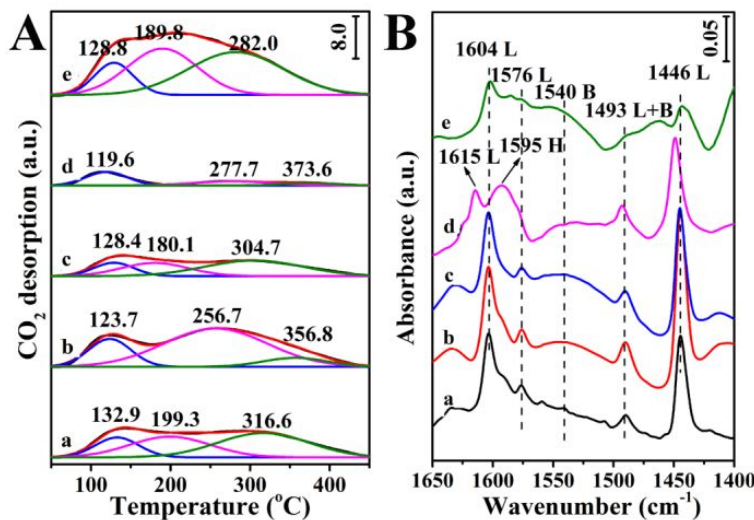


Figure 3. (A) CO₂-TPD profiles and (B) FT-IR spectra after pyridine adsorption of (a) Ag/Mg₂Al-LDO, (b) Ag/Mg₃Al-LDO, (c) Ag/Mg₄Al-LDO, (d) Ag/Al₂O₃, (e) Ag/MgO, and (f) Mg₄Al-LDO.

Table 1. Specific surface area and acid-base properties of the supported Ag particles.

Samples	S _{BET} /m ² ·g ⁻¹	Basic sites/mmol·g ⁻¹				Basic density /μmol·m ²	Acidic sites/ mmol·g ⁻¹	Acidic density /μmol·m ²	Ratio of acid to base
		weak	medium	strong	total				
Ag/Mg ₂ Al-LDO	57.2	0.09	0.18	0.24	0.51	8.91	0.21	3.63	0.41
Ag/Mg ₃ Al-LDO	68.3	0.10	0.33	0.05	0.48	7.02	0.23	3.38	0.48
Ag/Mg ₄ Al-LDO	86.0	0.06	0.11	0.18	0.35	4.07	0.24	2.79	0.69
Ag/Al ₂ O ₃	71.8	0.08	0.03	0.02	0.13	1.82	0.32	4.41	2.42
Ag/MgO	174.4	0.09	0.23	0.29	0.61	3.50	0.02	0.10	0.07
Mg ₄ Al-LDO	85.0	0.09	0.15	0.13	0.37	4.35	0.24	2.81	0.65

The concentration of weak, medium-strong, and strong basic sites are calculated according to the results of CO₂-TPD and the deconvoluted TPD profile in the temperature region of 50-450 °C. The acidic sites are calculated according to the results of pyridine-FTIR.

Then Ag loaded Mg_xAl-LDO, MgO, and Al₂O₃ have been applied in the dehydrogenation coupling of ethanol under ambient pressure at 350 °C (**Table 2**). For comparison, Mg₄Al-LDO has also been evaluated. Only 9.3 % of ethanol conversion (9.4 h⁻¹ of STC) is obtained on Mg₄Al-LDO with a selectivity of 36.8 % to n-butanol. Ethyl ether has also been produced in a selectivity of 36.8 %. The introduction of Ag particles on Mg₄Al-LDO not only motivates the conversion of ethanol, but also obviously promotes the dehydrogenation of ethanol and favors the condensation of the produced acetaldehyde, thus suppressing the intermolecular dehydration

of ethanol to ethyl ether. Increasing the amount of basic sites or decreasing the amount of acidic sites by modulating the Mg/Al molar ratio, the selectivity to n-butanol increases while that for ethyl ether decreases obviously. The intermolecular dehydration of ethanol to ethyl ether can be totally suppressed on Ag/MgO, in which the amount of basic sites is much higher while there are almost no acidic sites. But the condensation of aldehyde has also been passivated on Ag/MgO, and a selectivity of up to 65.1 % to acetaldehyde is obtained with only 29.7 % selectivity to n-butanol, implying that the acidic sites could promote the condensation of acetaldehyde. However, on Ag/Al₂O₃, which possesses only acidic sites, ethyl ether is observed as the dominant product, while almost no dehydrogenation of ethanol has occurred. The above results indicate that the dehydrogenation coupling of ethanol to n-butanol requires not only the catalysis of Ag particles but also the catalysis of both acidic and basic sites. In addition, each catalyst exhibits superior stability in a 750 min on-stream evaluation (**Figure 4**). For better comparison, the dehydrogenation coupling of ethanol on Ag/Mg₂Al-LDO was carried out at 250 °C and 2 MPa, following the procedure reported in literature. An initial ethanol conversion of 53.2 % with a selectivity of 57.3 % to n-butanol was achieved and an ethanol conversion of 45.2 % with a selectivity of 60.3 % to n-butanol was achieved at steady-state (**Table 2**), which is the highest selectivity at ethanol conversion of > 45 % till now.

Table 2. Catalytic results for upgrading of ethanol on supported Ag particles. ^a

Catalyst	Con./%	STC ^b /h ⁻¹	Sel./mol %								C balance /mol %
			ethyl ether	acetald ehyde	ethyl acetate	butanal	n-BuOH	i-hexanol	n-hexanol	other s	
Mg ₄ Al-LDO	9.3	9.4 ^c	36.8	11.5	1.2	1.1	46.7	0.4	0.7	1.6	95.2
Ag/Mg ₄ Al-LDO	31.7	3009.3	24.2	6.5	0.5	0.4	62.2	1.0	1.6	3.7	85.7
Ag/Mg ₃ Al-LDO	28.5	2706.3	13.0	10.2	1.0	0.6	69.2	1.3	1.8	2.9	84.1

Ag/Mg ₂ Al-LDO	23.2 (23.1)	2200.8 (2188.5)	9.3 (9.9)	5.3 (4.9)	0.5 (0.5)	0.5 (0.7)	76.6 (75.6)	1.8 (1.5)	3.5 (3.0)	2.5 (3.9)	83.2 (82.8)
Ag/MgO	5.5	524.2	-	65.1	1.6	-	29.7	-	-	3.6	80.1
Ag/Al ₂ O ₃	93.5	10159.8	96.5	3.1	-	-	0.1	-	-	0.3	87.6
Ag/Mg ₂ Al-LDO ^d	53.2 (53.7)	4124.9 (4104.7)	18.4 (19.0)	3.5 (3.3)	1.9 (1.8)	2.3 (2.2)	57.3 (56.8)	7.5 (7.8)	4.9 (5.0)	4.2 (4.1)	91.3 (90.8)
Ag/Mg ₂ Al-LDO ^e	45.2 (45.8)	3728.5 (3900.8)	17.8 (18.1)	3.2 (2.8)	1.6 (1.3)	0.3 (0.2)	60.3 (62.6)	7.2 (8.1)	4.8 (5.5)	4.7 (4.9)	92.3 (91.8)

^a Conversion and selectivity were obtained at initial point; reaction conditions: 500 mg of catalyst, 350 °C, 60 mL min⁻¹ of N₂ (0.1 MPa), LHSV = 6 mL (h g cat)⁻¹. ^b STC for ethanol conversion are calculated by the mole of ethanol converted on per mole of surface Ag per hour. ^c STC for ethanol conversion is calculated by the mmole of ethanol converted on per gram of catalyst per hour. ^d Conversion and selectivity were obtained at initial point; reaction conditions: 250 °C, 2 MPa. ^e Conversion and selectivity were obtained at steady-state; reaction conditions: 250 °C, 2 MPa. Other products include butyl acetate, ethyl butyrate, 2-ethyl-butanal, hexanal, ethyl 2-ethyl butyrate, butyl butyrate, ethyl caproate, etc. The data in parentheses are the reproduced experimental data.

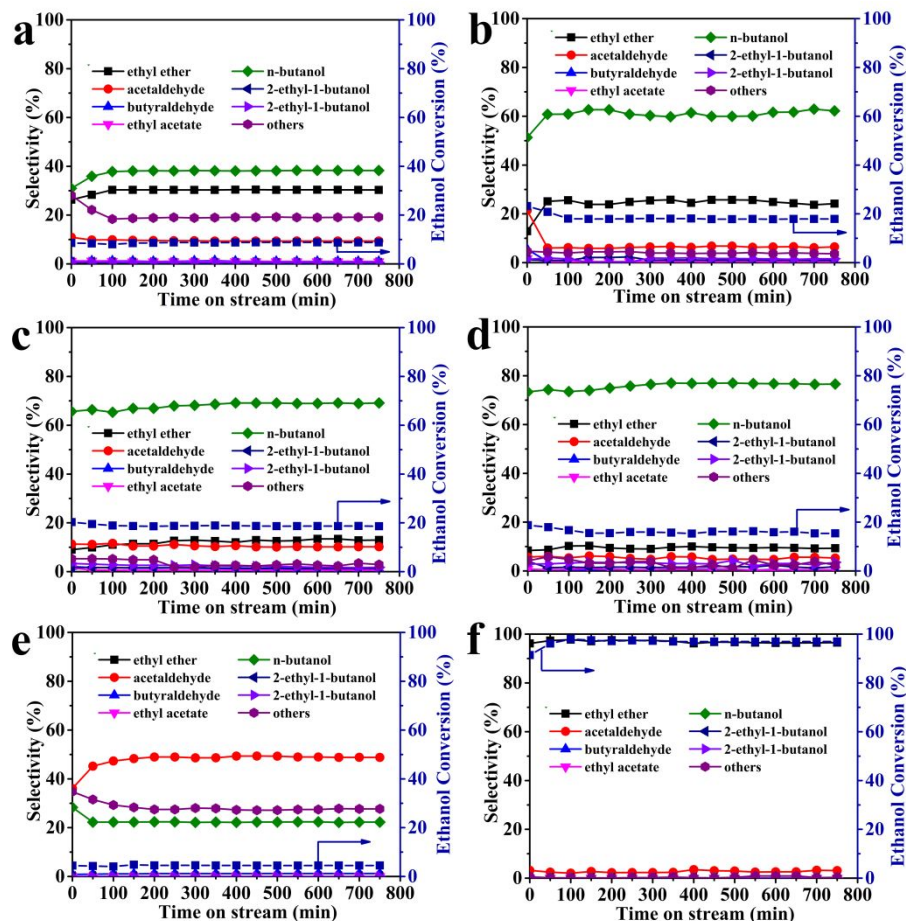


Figure 4. Stability of (a) Mg₄Al-LDO, (b) Ag/Mg₄Al-LDO, (c) Ag/Mg₃Al-LDO, (d) Ag/Mg₂Al-LDO, (e) Ag/MgO, and (f) Ag/Al₂O₃.

To investigate the roles of acidic or basic sites and Ag particles in the dehydrogenation coupling of ethanol to n-butanol, FT-IR spectra of ethanol adsorption at 50 °C followed by temperature-programmed desorption were recorded (**Figure 5a-d**). The ethanol adsorption at 50 °C on Ag/Mg₄Al-LDO (**Figure 5a**) presents absorption bands at 2987, 2968, 2927, 2901, 2868, 1617, 1451, 1393, 1250, 1128, 1101, and 1056 cm⁻¹. According to a previous report⁶², the bands at 2968, 2927, 2901, and 2868 cm⁻¹ can be assigned to the stretching modes of the CH₃ group and CH₂ group in adsorbed ethoxide, the band at 2987 cm⁻¹ to the stretching modes of the CH₃ group in adsorbed acetaldehyde, the band at 1617 cm⁻¹ to η¹-adsorbed acetaldehyde, the bands at 1451

and 1393 cm^{-1} to δ modes of CH_2 and CH_3 group, the band at 1250 cm^{-1} to δ modes of OH in undissociated adsorbed ethanol, the band at 1128 cm^{-1} to C-C in ethanol or acetaldehyde, and the bands at 1101 and 1056 cm^{-1} to the C-O stretching in adsorbed ethoxide. These results evidence that the activation of ethanol on Ag/Mg₄Al-LDO has occurred through the activation of O-H bond, affording ethoxide on Lewis sites and hydrogen on base sites, followed by dehydrogenation to acetaldehyde. With increasing temperature to desorption, the absorption at 50 °C decreases and vanishes above 200 °C while the band for η^1 -adsorbed acetaldehyde slightly increases. On Mg₄Al-LDO (**Figure 5b**), similar results are observed except that no bands at 2987 and 1617 cm^{-1} for the acetaldehyde are observed at 50 °C, indicative of the crucial role of Ag particles in the dehydrogenation of ethoxide to acetaldehyde. But two new bands at 1583 and 1443 cm^{-1} , assigned to acetate,⁶² are observed above 350 °C, indicating that the dehydrogenation could occur on basic sites (without Ag) but requires higher temperature. On MgO (**Figure 5c**), the bands for acetate are observed above 150 °C and the band for η^1 -adsorbed acetaldehyde is observed above 250 °C, confirming the promotion of basic sites on the dehydrogenation. On Al₂O₃ (**Figure 5d**), an absorption at 1169 cm^{-1} , assigned to the C-C bond of diethyl ether,⁶³ is visibly observed, which is consistent with the catalytic result (**Table 2**) that intermolecular dehydration dominates. The above results evidence the vital role of Ag particles in the dehydrogenation of adsorbed ethoxide to acetaldehyde and the promotion of basic sites to the dehydrogenation on Ag particles. The increase in the selectivity to dehydrogenation products, or products other than diethyl ether, with increasing amount of basic sites (**Figure 5e**) or decreasing acidic sites (**Figure 5f**) further demonstrate the promoting role of basic sites.

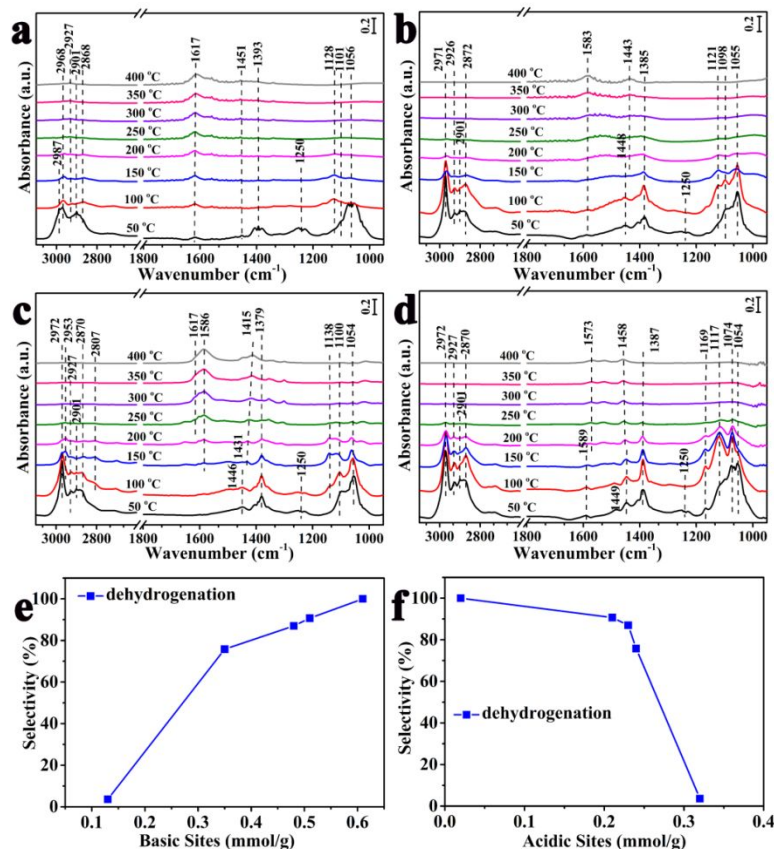


Figure 5. FT-IR spectra of ethanol adsorption on (a) Ag/Mg₄Al-LDO, (b) Mg₄Al-LDO, (c) MgO, and (d) Al₂O₃ at 50 °C followed by desorption at 100 °C, 150 °C, 200 °C, 250 °C, 300 °C, 350 °C, and 400 °C. Selectivity of dehydrogenation as a function of (e) basic sites and (f) acidic sites.

The aldol condensation of acetaldehyde is a crucial step for the C-C coupling, giving C₄ products. According to previous reports, the condensation can proceed over both basic sites^{64, 65} and acidic sites^{66, 67}. To investigate the acetaldehyde condensation, the FT-IR spectra of acetaldehyde adsorption at 10 °C have been recorded (**Figure 6A**). On Ag/Mg₄Al-LDO, the acetaldehyde adsorption gives 1730 and 1700 cm⁻¹ assigned to $\nu(\text{C}=\text{O})$ ⁶⁸, 1463 cm⁻¹ to $\delta_{\text{as}}(\text{CH}_3)$ ⁶⁸, 1398 cm⁻¹ to $\delta_{\text{s}}(\text{CH}_3)$ ⁶⁸, and 1582 cm⁻¹ to $\nu_{\text{as}}(\text{OCO})$ of acetate⁶⁸. Besides the above absorption attributed to acetaldehyde and acetate, the bands at 1650 cm⁻¹ attributed to $\nu(\text{C}=\text{O})$ of crotonaldehyde or 3-hydroxybutanal⁶⁸, 1611 cm⁻¹ to $\nu(\text{C}=\text{C})$ of enolate⁶⁸, 1337 cm⁻¹ to the

$\delta_{\text{as}}(\text{CH})$ of crotonaldehyde⁶⁸, and 1278 cm^{-1} to the $\delta(\text{C-OH})$ in adsorbed 3-hydroxybutanal⁶⁸ are also observed. On $\text{Mg}_4\text{Al-LDO}$ (**Figure 6A, b**), the band for $\delta_{\text{s}}(\text{CH}_3)$ is red-shifted to 1378 cm^{-1} and the bands for $\nu(\text{C=O})$ and $\delta_{\text{as}}(\text{CH})$ of crotonaldehyde become less dominant. That indicates that the Ag particles could facilitate the activation of C-H of CH_3 in acetaldehyde to promote the aldol condensation. For comparison with MgAl-LDO , the acetaldehyde adsorption on Al_2O_3 and MgO has also been recorded. On Al_2O_3 (**Figure 6A, c**), almost no bands for $\nu(\text{C=O})$ of crotonaldehyde, $\nu(\text{C=C})$ of enolate, and $\delta_{\text{as}}(\text{CH})$ of crotonaldehyde are observed, indicative of the poor aldol condensation on the singly acidic sites. On MgO (**Figure 6A, d**), the bands for $\nu(\text{C=O})$ of crotonaldehyde, $\nu(\text{C=C})$ of enolate are less dominant than that on $\text{Mg}_4\text{Al-LDO}$. As a consequence, almost no C-C coupling products, including butanal, n-butanol, hexanol, and butyl acetate, has been obtained with higher acid/base ratio (**Figure 6B-D**), and lower selectivity to C-C coupling products has been achieved with lower acid/base ratio. Higher selectivity to C-C coupling products has been achieved with appropriate amount of acidic sites and basic sites, and ratio of acid to base, indicative of the synergistic catalysis between acidic and basic sites in aldol condensation.

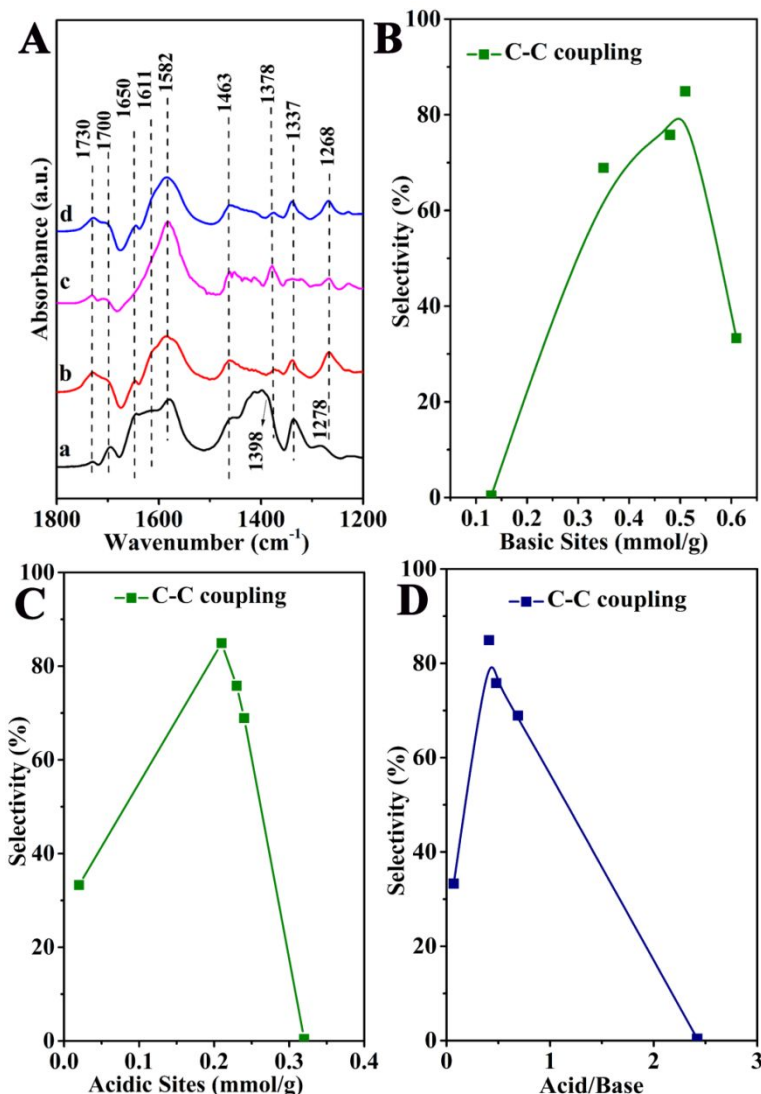


Figure 6. (A) FT-IR spectra of acetaldehyde adsorption on (a) Ag/Mg₄Al-LDO, (b) Mg₄Al-LDO, (c) Al₂O₃, and (d) MgO. Selectivity of the C-C coupling product as a function of (B) basic sites, (C) acidic sites, and (D) the ratio of acidic sites to basic sites.

4. CONCLUSIONS

In summary, LDO-supported Ag particles have demonstrated synergistic catalysis in the dehydrogenation coupling of ethanol. Up to 76.6 % of selectivity toward n-butanol with ethanol conversion of 23.2 % at 350 °C under ambient pressure has been achieved. Basic sites promote

ethanol dehydrogenation on Ag particles and acid-base cooperation promotes the following aldol condensation. The Ag particles have been found to play a vital role not only in ethanol dehydrogenation but also in aldol condensation. So the effects of particle size or electronic states of LDO-supported Ag in the dehydrogenation coupling of ethanol deserve more attention and investigation.

ASSOCIATED CONTENT

Supporting Information.

Figure S1-S18: GC-MS data of the products. This information is available free of charge via the Internet at <http://pubs.acs.org>.

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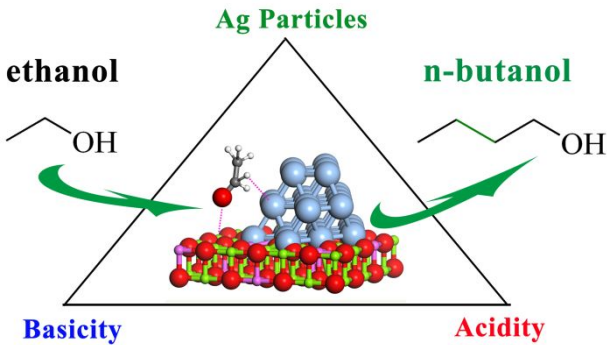
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Dehydrogenation coupling of ethanol to higher alcohol is achieved by the Ag-base-acid sites catalysis.